Mitchell's Principle of Constraint-First Unification: relations, never values

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"Don't ask a complex system for its values — ask for its constraints, its count, and its deficit. Everything else is covariance, not knowledge."

Abstract

The unification of string theory is not a formula for the Standard Model's parameters; it is a complete characterization. We state, and assemble the evidence for, a single principle: a theory of everything predicts *relations*, never *values*. The exact content of the flux sector is exhausted by one constraint — the D3 tadpole, the unique duality invariant *per symplectic pair* (a second independent sector brings its own cross-tadpoles) — together with the topological labels and the counting (nineteen, the Standard Model orbit-space dimension). Everything else transforms by a covariance table, Mitchell's Laws of Duality Covariance, measured on supersymmetric vacua at $h^{1,2} = 2$ and $h^{1,2} = 5$. And there is no selection: not exact (a no-go theorem), not statistical (the flux measure is flat), not cosmological (the cosmological constant is locally flat, and the uplift that would supply it is contested). The principle reduces to one equation: the invariant ring of the flux is $\mathbb{C}[N]$ with $N = F^\top J H$. We state the four falsifiers that would kill it.

1 The principle, in one equation

The whole of the programme follows from a single line of invariant theory. Let (F, H) be the type IIB three-form flux pair, valued in $H^3(X, \mathbb{Z}) \times H^3(X, \mathbb{Z})$, and let J be the symplectic intersection form. The duality group $\mathrm{Sp}(b_3, \mathbb{Z}) \times \mathrm{SL}(2, \mathbb{Z})$ acts on the pair, and the First Fundamental Theorem of invariant theory (Weyl 1939), specialized to two vectors, gives

$$\mathbb{C}[F, H]^{\,\mathrm{Sp}(b_3,\mathbb{Z})\times\mathrm{SL}(2,\mathbb{Z})} \;=\; \mathbb{C}[N], \qquad N = F^\top J\, H. \tag{1}$$

The ring of *exact, duality-invariant* quantities in the flux is generated by a single quadratic invariant — the D3 tadpole. Everything this note reports is a consequence of Eq. (1), stated in three tiers and never confused: **proven** (symbolic or machine-checked), **measured** (numerical, with a control that could have failed), and **conjectured** (open).

Principle 1 (Constraint-First Unification)**.** *A theory of everything predicts relations, never values. The exact content of a sector is exhausted by three types — constraints, labels, and counting. Everything else transforms by a covariance table. And there is no selection: not exact, not statistical, not cosmological.*

2 The exact structure is constraint + label + counting

The theorem tower assembles Eq. (1) and its consequences; the full proofs and the symbolic verification are in the companion paper *Constraint-First Theorem* (cited as paper 1). The chain is:

- **Mitchell's Uniqueness.** $N = F^\top JH$ is the *unique* bilinear symplectic invariant of the flux, for any compactification: $F^\top F$, $H^\top H$, $F^\top H$ are gauge-dependent, and $F^\top JF = H^\top JH = 0$.
- **Constraint-First.** For a single flux pair, every duality-invariant exact polynomial in (F, H) is a power of N — the invariant ring is generated by the tadpole alone (Weyl's first fundamental theorem at $k = 2$). *Scope:* with k independent flux pairs the invariant bilinear forms are the k^2 pairings $F_i^\top JH_j$, so a compactification with two independent sectors has four exact observables rather than one.
- **Completeness.** Over $\mathbb{R}$ the duality group acts transitively on each nonzero tadpole level set; over $\mathbb{Z}$ the orbits are classified by the content $d = \gcd(F, H)$ with $d^2 \mid N$, finitely many per tadpole.
- **Full-duality.** The statement holds for the full group $\mathrm{Sp}(b_3, \mathbb{Z}) \times \mathrm{SL}(2, \mathbb{Z})$, S- and T-duality included.
- **Complete Exact Structure.** The exact content is *exhausted* by constraints (the tadpole $N \leq L$, flux quantization, RR tadpoles, anomaly cancellation) and labels (the topology $h^{1,1}, h^{2,1}, c_2, \kappa$). The superpotential W_0, the string coupling g_s, the masses, the couplings, and the cosmological constant are provably *not* exact observables.

The immediate consequence is a no-go: *no exact, polynomial, duality-invariant selector can single out a vacuum.* This rules out the equation; it does not rule out statistical, moduli-dependent, or non-polynomial selection — a distinction the later sections take up.

3 The covariance table

The theorem says what is invariant; the laws say what is covariant. Measured on 100 primitive supersymmetric flux vacua at $h^{1,2} = 2$ and 12 in-cone supersymmetric vacua at $h^{1,2} = 5$, every measurement gated by a control, the transformation of the surviving observables under $\mathrm{SL}(2, \mathbb{Z})$ is:

law	content	$h^{1,2} = 5$
I (scaling)	$W(dF, dH) = d\,W$, $N \to d^2 N$	symbolic
II (magnitude)	$\lvert W_S\rvert/\lvert W\rvert = 1$	1.000000 (std 3.7×10^{-15})
IIb (phase)	$\arg(W_S/W) = -\arg\tau$	residual 7.4×10^{-16}
III (g_s)	$g_s \to \lvert\tau\rvert^2 g_s$	1.000000 (std 1.8×10^{-16})
IV (moduli)	complex-structure moduli inert at degree 0	—
V (counting)	19 = SM orbit-space dimension	—

These are the textbook type IIB S-duality laws, exactly. The laws are $h^{1,2}$-independent: the covariance table does not scatter as moduli are added. In parallel, the supersymmetric-vacuum fraction collapses with the number of moduli kept — the in-cone fraction falls $80\% \to 25\% \to 0\%$ from $h^{1,2} = 2$ to 5 to 10, while the F-flat points still exist. The physical vacua move *out* of the Kähler cone; the exact structure does not move. The collapse is a landscape fact, not a proposal artefact.

4 The Standard-Model frontier

One sector up, the same decomposition holds. The Standard Model's exact content is constraints (anomaly cancellation, proven 5/5; hypercharge uniqueness, proven 4/4), labels (the gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, $N_{\mathrm{gen}} = |\chi|/2$ a topological index), and parameters. The flavor group $\mathrm{U}(3)^5$ leaves an orbit space of dimension nineteen (Molien–Weyl), matching the literature count exactly: six quark masses, three mixing angles, one phase, three charged-lepton masses, three gauge couplings, two Higgs parameters, and θ_{QCD}. Ultraviolet completions do impose exact relations — minimal SU(5) collapses five (gauge 2 plus the three $Y_e = Y_d^\top$ mass relations $m_e = m_d$, $m_\mu = m_s$, $m_\tau = m_b$), and minimal SO(10), E_6, Pati–Salam, and trinification collapse twelve ($Y_u = Y_d = Y_e$, $V_{\mathrm{CKM}} = 1$) — but those relations are *numerically wrong* against data, so only gauge-coupling unification (two) survives. The nineteen are free, not derived.

The honest boundary: "the nineteen are free within the Standard Model" is a counting fact (proven); "no ultraviolet completion derives the values correctly" is a conjecture — the flux theorem worked because the flux has a clean duality group and a first fundamental theorem, which the Standard Model couplings lack. The surviving, defensible statement is: no unique *value* is derived — only *relations*, and the relations that are known (gauge unification, Yukawa unification) are mostly wrong against data, leaving two.

5 The universal-structure catalog

An earlier, stronger guess — that the sector-spanning content is *one* object (one K-theory vanishing, one coupling, one observable) — was tested by three independent literature passes and *refuted*. What survives is a small catalog:

1. **two kinds of universal constraint** — topological/index vanishing (the tadpole and the anomaly, the same kind via Atiyah–Singer and anomaly inflow) and positivity/analyticity (unitarity and causality);
2. **two universal couplings** — gravity (the unique coupling to the energy-momentum tensor) and the dilaton/axion (universal to $\mathrm{Tr}\, F^2$);
3. **the channel between flux and the Standard Model is the moduli** — $1/g^2 = \mathrm{Re}\, f(S, T, U)$, the gauge kinetic function, not gravity.

One line: a theory of everything is not one object; it is one *structure* — constraint, label, counting — repeated in every sector, wired together by the moduli.

6 No selection

The "why this vacuum" question has three levels, and each is answered negatively.

Exact. No invariant selector exists — this is the theorem of Section 2. **Statistical.** The flux measure is flat: measured on 60 supersymmetric vacua with the physical tadpole bound, $\arg W_0$ is uniform ($\langle\cos\rangle = 0.015$, $\langle\sin\rangle = -0.112$), reproducing the Denef–Douglas uniformity of W_0. No peak, no selection. **Cosmological.** The pre-uplift scaling variable $|W_0|^2$ has constant density near zero (measured: the cumulative fraction grows quadratically), so the cosmological constant is the locally-flat tail of a discretuum — not a peak. The uplift that would supply a small positive cosmological constant is *contested*: a 2022–2026 control-problem wave has now hit every uplift

mechanism, and the de Sitter swampland is strongly supported at the ten-dimensional level, though it is not a theorem. The field is trending toward "forbidden in the controllable regime," with no no-go established.

There is therefore no selection anywhere we can reach: not exact, not statistical, not cosmological.

7 The two open gates

Two items remain open, and they are stated as open, not claimed.

1. **The Standard-Model extension (conjectured).** That the nineteen are free in any ultraviolet completion, beyond the proven within-SM counting, is conjectured. The falsifier is concrete: any ultraviolet relation among the nineteen beyond GUT's three breaks it.

2. **The de Sitter uplift (the community's).** Every uplift mechanism now fails the ten-dimensional control test; the decisive question — *is the control problem generic?* — is string phenomenology's hardest open problem, and this note characterizes it rather than pretending to close it.

8 Falsifiers

The principle is a claim because it can be killed. Four observations would do it.

1. a moduli-independent, duality-invariant flux quantity that is *not* a function of $F^\top JH$ — breaks the ring $\mathbb{C}[N]$;

2. an exact, derivable, *and numerically-correct* relation among the nineteen beyond gauge-coupling unification (two) — breaks the counting;

3. a covariance law that *scatters* at higher $h^{1,2}$ — breaks the covariance table;

4. a measured non-uniform $dN/d|W_0|^2$ near zero, or a proven de Sitter no-go that empties the positive-Λ discretuum — breaks the no-selection statement.

9 Conclusion

Unification, correctly stated, is not a formula and not one object. It is a complete characterization: one constraint, a count, a covariance table, and the absence of selection. That characterization is proven for the flux, measured for the laws and the statistics, and conjectured — openly — for the Standard Model. A theory of everything predicts relations, never values.